
Instructions. (15 points) Sections 16.7, 16.8, 16.9. Show relevant work for full credit.

(3^{pts}) **1.** Compute $\iint_S z \, dS$ where S is the cylindrical surface $x^2 + y^2 = 1$, $0 \leq z \leq 2$.

(3^{pts}) **2.** Compute $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F}(x, y, z) = \langle 0, 0, z \rangle$ and S is the part of the paraboloid $z = 4 - x^2 - y^2$ lying above the xy -plane, oriented *upward*.

(3^{pts}) **3.** Let $\mathbf{F}(x, y, z) = \langle z^2, 2x, -y^3 \rangle$ and let C be the unit circle $x^2 + y^2 = 1$ in the plane $z = 0$, oriented **clockwise** when viewed from above. Use Stokes' Theorem to compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$.

(3^{pts}) **4.** Let $\mathbf{F}(x, y, z) = \langle xz^2, y^3, x^2z \rangle$ and let S be the unit sphere $x^2 + y^2 + z^2 = 1$ with outward orientation. Use the Divergence Theorem to compute $\iint_S \mathbf{F} \cdot d\mathbf{S}$.

- (3^{pts}) 5. For each of the following, identify the single *best* method for evaluation: **direct parametrization**, **FTLI**, **Green's Theorem**, **Stokes' Theorem**, or the **Divergence Theorem**. Give a one-sentence reason. (You do not need to evaluate any of them.)

(a) $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = \langle xy, e^{x^2}, z \rangle$ and C is the triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ traversed once.

(b) $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F} = \langle x + e^y, y + \sin z, z \rangle$ and S is the boundary of the cube $[0, 1]^3$ with outward orientation.

(c) $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = \langle 2x, 2y, 2z \rangle$ and C is any smooth path from $(0, 0, 0)$ to $(1, 2, 3)$.